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Using Scenario Modeling to Estimate Burden of Human Parvovirus B19 and Impact of Increased B19 Detection in Pregnancy

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ABSTRACT

Context: Human parvovirus B19 (B19) is a common infection that typically causes mild symptoms but can cause severe fetal complications among pregnant individuals. There are limited data to inform public health responses because B19 is not a reportable disease.

Objective: We sought to estimate the burden of B19-related pregnancy complications and impact of increasing B19 detection in both Minnesota and the United States using electronic health record data and scenario modeling.

Design: To estimate the burden of B19-related complications for the United States (4 000 000 annual pregnancies), a spreadsheet-based model was developed. B19 outcomes were estimated under varying scenarios of population immunity (25%–75%) and probability of infection (1%–7.5%). To evaluate the impact of increased B19 detection in pregnancy, a decision tree model compared the status quo (5% detection of cases) to a strategy that increased detection (50%). Inputs for both models were derived from published literature or expert opinion and were varied in sensitivity analyses.

Setting: Not applicable.

Participants: Not applicable.

Intervention: Increasing detection of B19 infections during pregnancy.

Main Outcome Measures: Main outcomes included number of stillbirths and intrauterine transfusions.

Results: For the United States nationally, the model estimated 750–2250 B19-related fetal complications annually under scenarios of 1% infection probability and 5625–16,875 with 7.5% infection probability, depending on proportion of pregnant individuals immune. Increasing the percent of maternal infections detected early from 5% to 50% could avert 42 stillbirths and lead to 35 more intrauterine transfusions under the 1% infection scenario. With the 7.5% infection scenario, 50% detection could avert 318 stillbirths and lead to 265 more transfusions annually.

Conclusions: In the context of limited B19 data, this analysis aimed to support public health investigation and response by providing estimates of B19 burden in pregnancy and evaluating the potential impact of increasing early detection of B19.

KEY WORDS: electronic health records, epidemiology, parvovirus B19, public health, scenario modeling

Introduction

Human parvovirus B19 (B19) is a common pathogenic virus transmitted through respiratory droplets that typically causes asymptomatic infection or mild illness in healthy adults and children. However, infection during pregnancy can lead to complications, such as severe fetal anemia, hydrops fetalis, thrombocytopenia, miscarriage, and stillbirth.[1–3] The risk of adverse fetal outcomes is highest when B19 infections occur within the first 20 weeks of pregnancy.[4] If fetal complications arise before 16–18 weeks' gestation (WG), there is no viable treatment for severe fetal anemia as intrauterine transfusions cannot be conducted.[5] If

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severe fetal anemia or hydrops develop after 16–18 WG, it can be treated by intrauterine transfusion.[5] However, if severe fetal anemia is undetected or transfusions are performed too late, then fetuses may progress to stillbirth.

The American College of Obstetricians and Gynecologists (ACOG) currently recommends pregnant individuals be monitored for B19 if they have symptoms or a known exposure.[6] Because the majority of maternal B19 infections are asymptomatic or present with nonspecific symptoms, maternal cases are rarely identified or are identified only after fetal complications arise.[7,8] If a maternal infection is detected during pregnancy, the fetus is monitored closely with serial ultrasonography to check for the development of severe fetal anemia and fetal hydrops (fluid buildup in at least 2 fetal compartments) so that transfusions can be performed as soon as clinically indicated.[8]

There is no routine prenatal screening for B19 because it is not a nationally notifiable condition in the United States.[4] The number of pregnancies and severe fetal complications related to B19 is unknown. B19 transmission is known to be cyclical, with peaks in infection prevalence occurring approximately every 3–4 years.[7] The most recent uptick in B19 cases was observed in 2024 across Europe and the United States, triggering a Centers for Disease Control (CDC) Health Alert Network notification in August 2024.[3,4] This increase was supported by electronic health record (EHR) data from Minnesota, which found 19 B19-associated pregnancy complications within 60 days of a B19 diagnosis in a 10-month period in 2024, compared with only 28 complications in the prior 60-month period from 2019 to 2023.[9] However, identifying B19 outbreaks is challenging and often delayed because of the lack of surveillance, limiting the effectiveness of public health and clinical responses.[9] Public health responses are further challenged by scarce published literature on the potential effectiveness of B19 interventions.

To support public health responses for current and future B19 outbreaks, we estimated the expected number of B19-related complications in pregnancy under different epidemiological scenarios and the potential impact of increasing detection of B19 infection during pregnancy.

Methods

Overview

We first estimated B19-related pregnancy complications using statewide EHR data from Minnesota. These results were used to validate model-based estimates. We then developed a spreadsheet-based scenario model to estimate potential B19 burden during pregnancy under different epidemiological scenarios for the United States followed by developing a decision tree model to evaluate the impact of increasing detection of maternal B19 infections on stillbirths and intrauterine transfusions. Models are publicly available on GitHub: https://github.com/MADM-C/Scenario-Modeling-of-Human-Parvovirus-B19. This project fell under public health surveillance, not research, and therefore Institutional Review Board approval was not obtained.

EHR data analysis

Retrospective data from 10 of the largest health systems in Minnesota, accounting for approximately 90% of Minnesota's patient population, were obtained by the Minnesota Electronic Health Records Consortium.[10] Diagnosis and procedure codes were used to identify fetal complications, including hydrops fetalis (P83.2), fetal anemia/thrombocytopenia (PO36.8290), intrauterine transfusion (36 460), and stillbirth (P95). Each site extracted clinical encounters with relevant diagnostic and/or procedure codes from their EHR data between January 2019 and December 2023 for patients with female sex aged <45 years because there was no established method for accurately ascertaining pregnancy status in these data. Outcomes were aggregated and summarized as the total number of clinical encounters for each fetal complication and number of these clinical encounters that occurred within 60 days of a positive B19 test result. These Minnesota-based EHR data were used to

Availability of data and materials: *The EHR data used and analyzed in this study are not publicly available. Data used in the scenario models were obtained from literature review and expert opinion. These values can be found in Table 1.The GitHub repository containing the files and code for running the scenario models can be found here.*

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validate outcomes from the models, using a population size of 60 000 pregnancies, which is approximately 90% of the total estimated number of annual pregnancies in Minnesota.[11] It should be noted that some B19-related complications may by missed in the EHR data because B19 testing is not routinely done during pregnancy. Furthermore, miscarriages are not reliably captured in EHR data but likely account for a significant portion of B19-related complications.

Burden of disease scenario modeling

We developed a spreadsheet-based model to estimate the potential number of pregnant individuals susceptible to B19, number of susceptible pregnant individuals infected with B19, and total number of severe fetal complications occurring after a B19 infection. Although B19 infection can also pose risks to pregnant individuals with certain rare underlying conditions, such as sickle cell disease, we focused on complication outcomes in the fetus. Complications include cases ending in fetal demise, either miscarriage or stillbirth, and fetuses that were untreated or treated and experienced live births. Outcomes were calculated by multiplying a series of event probabilities, starting with a population size of the annual total number of pregnancies (Figure 1A). Event probabilities were derived from published literature and expert opinion (Table 1).[2,12–23] To estimate national US burden, a population of 4 million pregnancies was used.[24] Results were estimated for scenarios in which 1% of susceptible pregnant individuals were infected with B19 within 20 WG versus 7.5% infected. These infection levels reflect "normal" or mild years versus a hypothetical severe outbreak year based on limited historic data.[16–19] With limited data to indicate current immunity to B19, results were estimated under assumptions of 25%, 50%, and 75% of pregnant individuals having B19 immunity.[12–16] Finally, the model assumed severe fetal complications occurred in 7.5% of pregnancies if the B19 infection occurred within 20 WG based on observational study.[20] Model input values were varied in sensitivity analyses.

Increasing early B19 detection decision modeling

The burden of disease model was then extended into a decision tree model that estimated the potential impact of increasing early detection of maternal B19 infections before fetal complications arise (Figure 1B). The model compared 2 strategies: 5% of maternal B19 infections detected early (estimated status quo) and 50% of maternal B19 infections detected early (increased detection). The detection probability (5%) for the status quo was based on expert opinion founded on clinical practice on how often B19 cases are detected before fetal complications develop, which is relatively uncommon but can still occur through patient–provider conversations where exposures are disclosed (eg, a pregnant mother mentions another child with slapped cheek rash). The 50% probability of detection under the increased detection strategy was a hypothetical increase. We estimated the number of B19-related stillbirths the and number of intrauterine transfusions performed to treat severe anemia under these 2 strategies under different epidemiological scenarios.

It was assumed that both detected and undetected cases had an equal baseline risk (7.5%) of developing severe fetal complications. However, early detected cases were assumed to undergo regular monitoring and had a 95% likelihood of receiving an intrauterine transfusion if fetal anemia occurred, compared with an 80% likelihood among undetected cases. These assumptions were based on maternal–fetal medicine (MFM) experiences in practice: most cases of fetal anemia are detected at the routine 20-week ultrasound; however, fetal death may occur in some cases if the anemia was detected too late or if the pregnant person was not interacting with the healthcare system. It was also assumed that fetuses with severe anemia had a higher probability of fetal death: 28% if the infection was undetected (no transfusion) and 20% if the infection was detected/monitored (transfusion occurred) compared with 10% if severe fetal anemia did not develop (Table 1). B19 immunity, infection, and detection probabilities were varied in sensitivity analyses.

Results

EHR findings & model validation

Between 2019 and 2023 in Minnesota, there were on average 393 possible B19-related fetal complication encounters per year among females aged <45 years, with an average of 5.6 encounters occurring within 60 days of a positive B19 test result each year. The average number of stillbirth encounters within 60 days of a positive B19 test was <2.2 per year. Assuming 60 000 pregnancies, 50% maternal immunity, and 0.5%–2% probability of maternal infection, the model estimated 1.6–6.4 B19-related stillbirths per year, which aligned with EHR findings of 5.6 average encounters within 60 days of a positive B19 test result. Model-based estimates for the total number of severe fetal complications caused by B19 was 11–45 per year. This was higher than the EHR findings, which did not include miscarriages.

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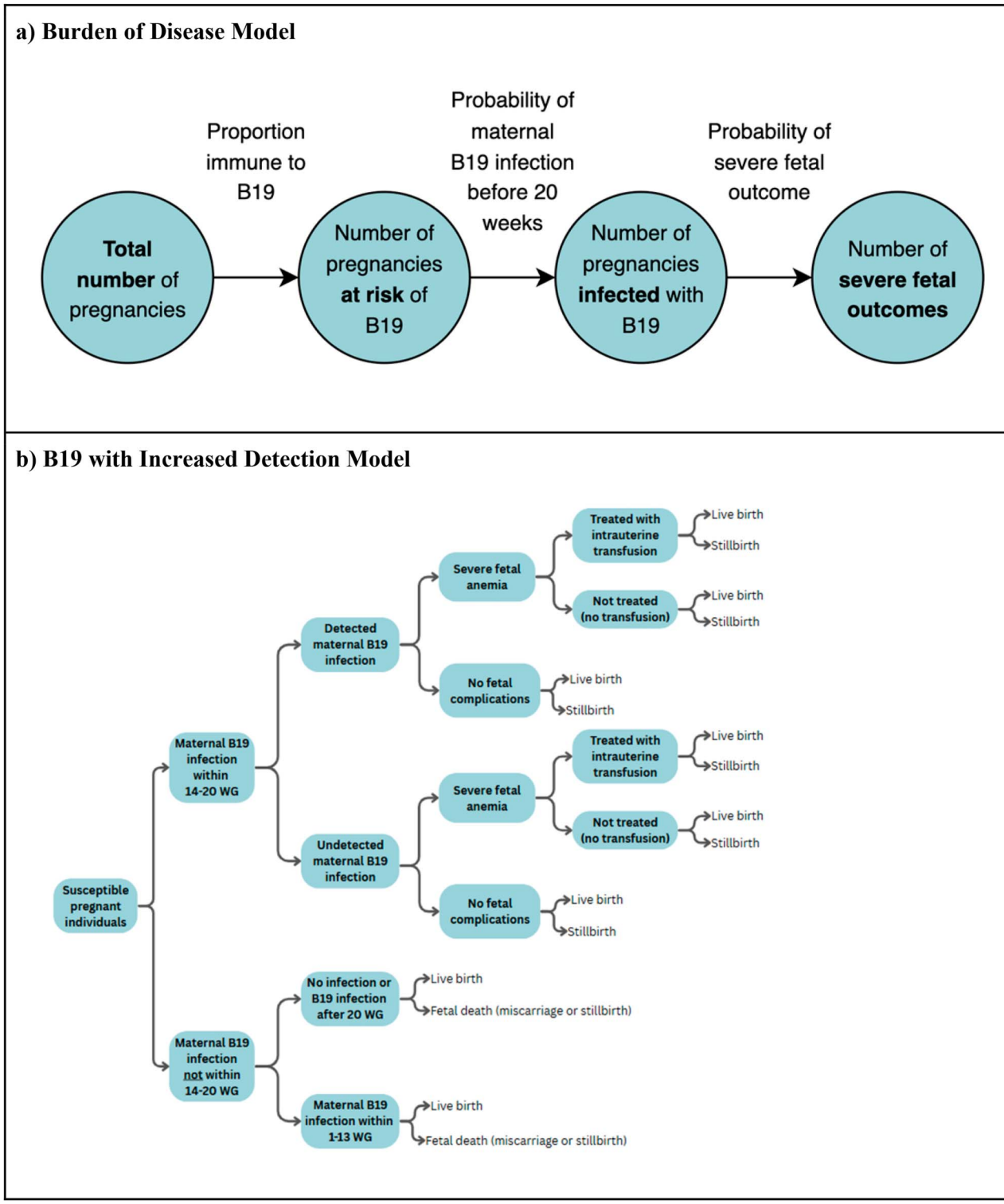


FIGURE 1 Model diagrams. A, The burden of disease model uses a simplified structure to estimate the total number of severe fetal complications based on event probabilities derived from published literature. It starts with the total number of pregnancies and calculates the number of pregnancies at risk of B19 infection, number infected with B19, and number of severe fetal complications resulting from B19. B, The increased detection model uses a decision tree to estimate the number of stillbirths and transfusions. Maternal B19 infections within 14–20 weeks' gestation (WG) can be detected early and have higher probabilities of transfusion and live birth if fetal anemia develops compared with undetected infections. Pregnancies that do not have a B19 infection within 14–20 WG experience normal live birth probability if there was no B19 infection or an infection after 20 WG when severe fetal complications are substantially less likely to develop. For maternal B19 infections within 1–13 WG, severe fetal complications that develop are considered untreatable and have a high probability of fetal death.

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TABLE 1
Model Parameter Values and Sources

Model Input Parameter	Base Case Value	Range in Sensitivity Analysis	Sources
Inputs used in both models			
Proportion of pregnant population immune to B19	50%	25%–75%	Ceccarelli, 2024; Crane, 2014; Lamont, 2010; Röhrer, 2008; Valeur-Jensen, 1999
Probability of maternal B19 infection within first 20 weeks' gestation if not immune			
During a mild transmission year	1%	0.5%–2%	Ergaz, 2006; Gessel, 2006; Gratacós, 1995; Valeur-Jensen, 1999
During a severe transmission year	7.5%	5%–10%	
Probability of severe fetal complications after maternal B19 infection	7.5%	5%–10%	Bonvicini, 2011; assumption based on clinical practice
Inputs used only in increased detection model			
Probability of detecting maternal B19 infection in status quo	5%	—	Assumptions based on clinical practice
Probability of detecting maternal B19 infection in increased detection strategy	50%	10%–50%	
Probability of intrauterine transfusion in undetected cases	80%	—	Assumptions based on clinical practice
Probability of intrauterine transfusion in detected cases	95%	—	
Probability of fetal death:			
No maternal B19 infection or maternal B19 infection with no fetal complications	10%	—	Andersen, 2000
Undetected maternal infection with fetal complications and no transfusion	28%	—	Kosian, 2022; assumption based on clinical practice
Detected maternal infection with fetal complications and transfusion	20%	—	
Proportion of B19-related fetal deaths that are miscarriages	78%[a]	—	Enders, 2004; Riipinen, 2008
Proportion of B19-related fetal deaths that are stillbirths	22%[a]	—	

[a]*Of 9 total fetal deaths, 7 were miscarriages (78%) and 2 were stillbirths (22%).*

Burden of disease

In a mild B19 year in the United States (1% infection probability in the first 20 WG), the estimated annual number of severe fetal complications ranged from 750 to 2250 (0.19–0.56 per 1000 pregnancies) depending on the proportion of pregnant individuals immune to B19 (Table 2). In a hypothetical severe outbreak year (7.5% infection probability), the estimated annual number of severe fetal complications may range from 5625 to 16 875 (1.41–4.22 per 1000 pregnancies), depending on the proportion of individuals immune.

Increasing early B19 detection

In a mild B19 year in the United States, increasing early detection of maternal B19 infections from 5% to 50% could avert an estimated 42 stillbirths and increase the number of performed intrauterine transfusions by 35, assuming 50% of pregnant individuals are immune to B19 (Table 3). In a hypothetical severe outbreak year,

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TABLE 2
Burden of Disease: Estimated Annual Number of B19-Related Severe Fetal Complications[a] in the United States

Burden of Disease Outcome	Mild B19 Transmission Year 1% Infection Probability			Severe B19 Transmission Year 7.5% Infection Probability		
	25% Immune	50% Immune	75% Immune	25% Immune	50% Immune	75% Immune
Pregnant individuals at risk of B19 (not immune)	3 000 000	2 000 000	1 000 000	3 000 000	2 000 000	1 000 000
Pregnant individuals infected with B19	30 000	20 000	10 000	225 000	150 000	75 000
Fetuses with severe complications because of B19	2250	1500	750	16 875	11 250	5625
Severe fetal complications per 1000 pregnancies	0.56	0.38	0.19	4.22	2.84	1.41

[a]*Severe fetal complications included estimated cases that ended in fetal demise, either miscarriage or stillbirth, and fetuses that were treated and experienced live births.*

increasing early detection could avert 318 stillbirths and increase performed transfusions by 265.

In 1-way sensitivity analyses, infection probability had the largest impact on B19-related stillbirths, followed by the proportion of pregnant individuals immune to B19 (Figure 2). In 2-way sensitivity analyses, the number of stillbirths averted with increased detection compared with status quo ranged from 24 to 636 as proportion of pregnant population immune varied between 75% and 25% and probability of infection varied from 0.5% to 10% (Supplementary Figure S1, http://links.lww.com/JPHMP/B710).

Discussion

As B19 is not a reportable disease and limited data are available to inform public health responses, this analysis sought to fill knowledge gaps by estimating B19-related complications in pregnancy using EHR data and scenario modeling. Compared with Minnesota-based EHR data that identified on average 5.6 fetal complications per year within 60 days of a positive B19 test result, model-based results estimated a higher burden (11–45 total severe fetal B19 complications per year) with the inclusion of miscarriage, an outcome not reliably captured in EHR data but which may account for the majority of B19-related fetal deaths.[22,23] Scenario modeling for the United States indicated that B19 may cause 750–2250 severe fetal complications in mild infection years and up to 5625–16,875 in severe outbreak years, depending on the proportion of pregnant individuals already immune to B19 (varied from 75% to 25%).

Our analysis supports current ACOG testing guidelines, given currently available diagnostics, and treatment options for fetal B19 infection. There are

TABLE 3
Estimated Impact of Increasing B19 Detection on Annual Number of Stillbirths and Intrauterine Transfusions in the United States

B19-Related Outcome	Mild B19 Transmission Year 1% Infection Probability		Severe B19 Transmission Year 7.5% Infection Probability	
	Status Quo (5% Detected)	Increased Detection (50% Detected)	Status Quo (5% Detected)	Increased Detection (50% Detected)
Total B19 stillbirths	214	172	1607	1289
B19 stillbirths per 1000 pregnancies	0.052	0.043	0.40	0.32
Total stillbirths averted	—	42	—	318
Total intrauterine transfusions performed	424	459	3180	3445
Increase in transfusions performed	—	+35	—	+265

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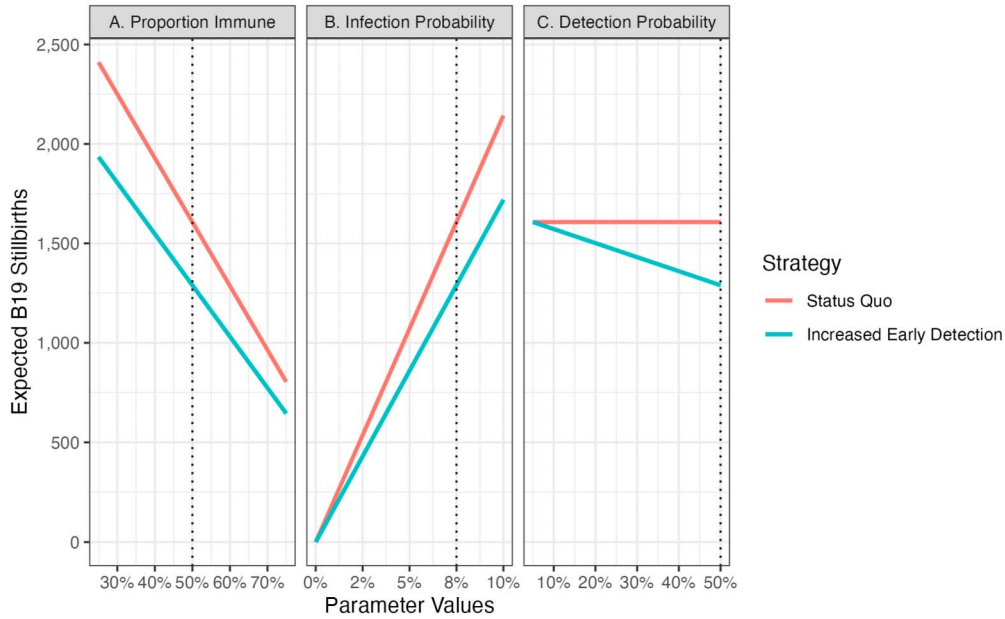


FIGURE 2 One-way sensitivity analyses for status quo versus increased early detection strategies. The 1-way sensitivity analysis demonstrates how the estimated number of B19-related stillbirths (out of 4 million pregnancies) is affected by probabilities of being immune, infected, and detected early. Dotted lines represent base case values. A, As immunity to B19 increases, B19 stillbirths decrease in both strategies. B, As maternal infection probability increases, B19 stillbirths increase most rapidly in the status quo strategy. C, B19 stillbirths remain constant in the status quo strategy but decrease steadily in the early detection strategy as the probability of detection increases.

approximately 21 000 stillbirths per year in the United States or 6 per 1000 births.[25] Our estimates indicate that increased B19 testing and monitoring may avert between 0.2% and 1.5% of stillbirths nationally, in low and high burden years, respectively. However, achieving this 50% early detection of maternal B19 infections would require much more widespread awareness of and testing for B19 during pregnancy. Increased testing and monitoring of pregnant individuals and their fetuses would also substantially increase testing and fetal monitoring costs and identifying maternal infections early and having limited intervention options may not be practical. Because the clinical window to treat a fetus with severe anemia is limited as transfusions can only be performed after approximately 18 WG, there will be many identified cases for which nothing can be done but wait. It may lead to healthcare overuse and contribute to increased stress and anxiety among pregnant individuals and their families. A formal cost-effectiveness analysis could further examine the costs, benefits, and potential harms associated with a more intensive clinical approach to identifying and managing B19 in pregnancy. Further study is warranted regarding the feasibility of the implementation of interventions during high prevalence seasons. The results of the modeling were shared with the Minnesota Department of Health (MDH) and the CDC. A health advisory was distributed by MDH to inform providers of higher levels of B19 circulating in the state. Providers may not usually suspect B19 as a cause for fetal hydrops, so alerting them to be aware of B19 is key for earlier interventions.

A routine surveillance system for B19 and the impact on pregnancies and fetuses would be valuable; however, funding and implementation may be more feasible if an effective public health and clinical response can be mounted. A surveillance system would support public health awareness of B19 trends and allow earlier identification of high prevalence years. In B19 outbreak years, clinicians could consider B19 testing in cases of pregnant individuals with flu-like symptoms and provide more education to pregnant individuals about it, so they are more likely to report exposures. A targeted testing program among pregnant individuals with higher risk of B19 exposure, such as daycare providers, in association with education on prevention measures such as handwashing during outbreak years could also be considered. Even moderately increasing detection of B19 cases from 5% to 20% could avert stillbirths when transfusions are possible. In the setting of a lack of a national or statewide surveillance system for B19, EHR data could be used to assist public health in assessing trends and B19 outbreaks or high prevalence periods by using natural language processing (NLP) of clinician notes.

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Such interventions would be collaborative efforts between public health, healthcare, and researchers. Surveillance from public health would guide healthcare decision-making about testing, and in the absence of data, researchers can use modeling to provide estimates.

A health equity consideration related to B19 burden in pregnancy is the concern for access to MFM specialist care to monitor for severe anemia and perform intrauterine transfusions, as needed. Access to MFM providers is limited in the United States, currently 35% of counties in the United States do not have access to obstetric care creating maternity deserts.[26] During a B19 outbreak, when more cases requiring intrauterine transfusions are identified, resource and access constraints are an important consideration. Early detection can play a role in resource and access planning. Pregnant individuals in maternity deserts may disproportionately experience stillbirths because of lack of access, delays receiving specialized MFM care, or resource limitations. In the growing number of regions with limited or no local obstetrics care, identification of B19-affected pregnancies may be even more critical so that timely transfer can be arranged to more specialized settings should more advanced monitoring or an intrauterine transfusion become necessary.[26] Ensuring equitable access to interventions is a necessity in any approach.

Our analysis had several limitations. First, the EHR data likely undercount the number of B19-related fetal complications because B19 testing is not routine, and miscarriages are not well-represented in EHR data; however, the extent of it is unclear. Second, validating model results was challenging because of lack of published data on B19 and undercounting in the EHR data. Although our results for B19-related stillbirths aligned with EHR data, the total number of B19 severe complications could not be validated with real-world data. Third, the available literature that was used to inform model parameters was from 20+ years ago or from other countries that are not representative of the US population. Our models therefore relied heavily on assumptions and clinical expertise. We also validated the model against EHR data from Minnesota, which may not be representative of the entire US population. We conducted sensitivity analyses on parameters that were dependent on assumptions or clinical expertise to help understand what parameters had the greatest impact on model results. Finally, simplifying assumptions were made for the analysis on the potential impact of increased detection of maternal infection: it was assumed that only maternal infections that occurred between 14 and 20 WG were intervenable and maternal infections <13 WG were excluded from our estimates on stillbirth.

Implications for Policy & Practice

- Health departments and health systems can use scenario modeling to evaluate changing disease trends, especially in situations with sparse data.
- Scenario modeling can be used to assess guidelines and ensure those guidelines are relevant and effective.
- Considerations for B19 surveillance could include enhanced EHR case detection or using NLP to monitor B19-related symptoms and key words.
- Policy what would improve equitable healthcare access for pregnant individuals in maternity deserts is a necessity.

Conclusion

In the context of limited data and literature on B19, this analysis aimed to support public health investigation and response. Our analysis is intended to inform outbreak response and increased detection strategies, not universal screening. Scenario modeling provided estimates of B19 burden and the potential benefits of increased detection. Given the lack of public health surveillance systems for B19, enhanced case detection using EHR data has the potential to prevent stillbirths through earlier detection and awareness of B19 outbreaks.

References

1. Heegaard ED, Brown KE. Human parvovirus B19. *Clin Microbiol Rev*. 2002;15(3):485-505. doi:10.1128/CMR.15.3.485-505.2002
2. Kosian P, Hellmund A, Geipel A, et al. Intrauterine transfusion in 103 fetuses with severe anemia caused by parvovirus infection. A multicenter retrospective study. *Arch Gynecol Obstet*. 2023;308(1):117-125. doi:10.1007/s00404-022-06712-z
3. Russcher A, Verweij E, Maurice P, et al. Extreme upsurge of parvovirus B19 resulting in severe fetal morbidity and mortality. *Lancet Infect Dis*. 2024;24(8):e475-e476. doi:10.1016/S1473-3099(24)00373-6
4. *Increase in Human Parvovirus B19 Activity in the United States*. 2024. Accessed January 23, 2025. https://www.cdc.gov/han/2024/han00514.html
5. Van Buren NL. Chapter 50—perinatal transfusion medicine. In: Shaz BH, Hillyer CD, Reyes Gil M, eds. *Transfusion Medicine and Hemostasis*. 3rd Edition. Elsevier; 2019:301-312. doi:10.1016/B978-0-12-813726-0.00050-7
6. CDC. *Increase in Human Parvovirus B19 Activity in the United States*; 2024. Accessed March 14, 2025. https://www.acog.org/clinical/clinical-guidance/practice-advisory/articles/2024/08/increase-in-human-parvovirus-b19-activity-in-the-united-states
7. Bennett JE, Dolin R, Blaser MJ. *Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases*. Elsevier Inc.; 2014. Accessed March 7, 2025. https://pure.johnshopkins.edu/en/publications/mandell-douglas-and-bennetts-principles-and-practice-of-infectiou

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8. *Red Book (2021): Report of the Committee on Infectious Diseases*. American Academy of Pediatrics; 2021. doi:10.1542/9781610025225
9. Contag S, Dufort EM, Lim S, et al. Notes from the field: human parvovirus B19 infections among pregnant persons—Minnesota, January-September 2024. *MMWR Morb Mortal Wkly Rep*. 2024;73(47):1087-1089. doi:10.15585/mmwr.mm7347a4
10. *Power BI Report*. Accessed July 31, 2025. https://app.powerbi.com/view?r=eyJrIjoiZjk1Yzc2ZmEtZjk0Ni00NTI5LWEzMWMtN2NkMjgxZmE3ZjYzIiwidCI6IjM4ZjIzNzIzLWM0ZmEtNDI1ZS04ZmE3LTFhNTk2NGUwNDIwYyIsImMiOjN9
11. Data by Topic—Births & Fertility. *MN State Demographic Center*. Accessed April 24, 2025. https://mn.gov/admin/demography/data-by-topic/births-fertility/
12. Ceccarelli G, Branda F, Ciccozzi A, et al. Reassessing the risk of severe parvovirus B19 infection in the immunocompetent population: a call for vigilance in the wake of resurgence. *Viruses*. 2024;16(9):1352. doi:10.3390/v16091352
13. Crane J, Mundle W, Boucoiran I, Maternal Fetal Medicine Committee. Parvovirus B19 infection in pregnancy. *J Obstet Gynaecol Can*. 2014;36(12):1107-1116. doi:10.1016/S1701-2163(15)30390-X
14. Lamont RF, Sobel J, Vaisbuch E, et al. Parvovirus B19 infection in human pregnancy. *BJOG*. 2011;118(2):175-186. doi:10.1111/j.1471-0528.2010.02749.x
15. Röhrer C, Gärtner B, Sauerbrei A, et al. Seroprevalence of parvovirus B19 in the German population. *Epidemiol Infect*. 2008;136(11):1564-1575. doi:10.1017/S0950268807009958
16. Valeur-Jensen AK, Pedersen CB, Westergaard T, et al. Risk factors for parvovirus B19 infection in pregnancy. *JAMA*. 1999;281(12):1099-1105. doi:10.1001/jama.281.12.1099
17. Ergaz Z, Ornoy A. Parvovirus B19 in pregnancy. *Reprod Toxicol*. 2006;21(4):421-435. doi:10.1016/j.reprotox.2005.01.006
18. van Gessel PH, Gaytant MA, Vossen ACTM, et al. Incidence of parvovirus B19 infection among an unselected population of pregnant women in the Netherlands: a prospective study. *Eur J Obstet Gynecol Reprod Biol*. 2006;128(1-2):46-49. doi:10.1016/j.ejogrb.2005.11.042
19. Gratacós E, Torres PJ, Vidal J, et al. The incidence of human parvovirus B19 infection during pregnancy and its impact on perinatal outcome. *J Infect Dis*. 1995;171(5):1360-1363. doi:10.1093/infdis/171.5.1360
20. Bonvicini F, Puccetti C, Salfi NCM, et al. Gestational and fetal outcomes in B19 maternal infection: a problem of diagnosis▽. *J Clin Microbiol*. 2011;49(10):3514-3518. doi:10.1128/JCM.00854-11
21. Nybo Andersen AM, Wohlfahrt J, Christens P, et al. Maternal age and fetal loss: population based register linkage study. *BMJ*. 2000;320(7251):1708-1712.
22. Enders M, Weidner A, Zoellner I, et al. Fetal morbidity and mortality after acute human parvovirus B19 infection in pregnancy: prospective evaluation of 1018 cases. *Prenat Diagn*. 2004;24(7):513-518. doi:10.1002/pd.940
23. Riipinen A, Väisänen E, Nuutila M, et al. Parvovirus B19 infection in fetal deaths. *Clin Infect Dis*. 2008;47(12):1519-1525. doi:10.1086/593190
24. *FastStats*. 2024. Accessed March 14, 2025. https://www.cdc.gov/nchs/fastats/births.htm
25. CDC. *Data and Statistics on Stillbirth*. Stillbirth; 2025. Accessed July 6, 2025. https://www.cdc.gov/stillbirth/data-research/index.html
26. Human Rights Research Center. *Maternity Deserts in the United States and the Threat to Women's Health*. HRRC; 2025. Accessed July 6, 2025. https://www.humanrightsresearch.org/post/maternity-deserts-in-the-united-states-and-the-threat-to-women-s-health

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